

Lim Xin Yan

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Predicted First-Class MEng Computer Science graduate from the University of Bristol (June 2026), with hands-on experience in deep learning research, full-stack development, and award-winning product engineering. Proven expertise in designing backend architectures, complex PostgreSQL schemas, and state-of-the-art low-light video enhancement systems, including a £5,000 award-winning project.

Education

University of Bristol, MEng Computer Science **Sept 2022 – June 2026**

- Predicted: First Class Honours. Recipient of the prestigious Think Big Scholarship, awarded to high-achieving international students. Earned the Bristol PLUS Award for completing accredited work experience alongside academic study.

INTI International College Penang, A-Levels **Apr 2021 – June 2022**

Mathematics (A*), Further Maths (A*), Chemistry (A*), Physics (A*)

Experience

Software Developer Intern, University of Bristol **May 2025 – Aug 2025**

- Investigated recursive modifications to the BVI-Mamba state-space model for low-light video enhancement, evaluating impacts on temporal stability and colour consistency.
- Applied findings to directly inform the curriculum learning approach developed in dissertation research, validated through large-scale training on HPC clusters.

Software Engineering Intern, Keysight Technologies Malaysia **June 2024 – Sept 2024**

- Automated the creation of driver folders based on order data, streamlining the workflow using WinForms (.NET), C# and SQL, reducing manual work by 80-90%.
- Developed an internal web tool adopted by the test engineering team to streamline software version verification, reducing dependency on manual checks.

Projects

Curriculum-Guided Recurrent Low-Light Video Enhancement **Jan 2026 – May 2026**

- Proposed a curriculum learning strategy for recurrent low-light video enhancement, progressively scaling degradation complexity across training stages to improve convergence.
- Adapted the ReMATF framework for low-light enhancement, integrating DINOv3 to leverage previous frames for improved temporal stability.
- Engineered a dynamic loss suite, achieving state-of-the-art average PSNR (30.27 dB) and outperforming the baseline ReMATF by +2.91 dB.
- **Technologies:** Python, PyTorch

DrugWise **June 2025 – Sept 2025**

- Winner of the University of Bristol Aegis Innovation Competition: Awarded £5000.
- Built a cross-platform web and mobile application providing accessible, evidence-based information on drug interactions, food interactions, and counterfeit medicine.
- Engineered the backend architecture by integrating distinct databases to power complex drug searches and implementing secure user authentication via Firebase Auth.
- **Technologies:** React, TypeScript, Supabase, Firebase Auth/Firestore, PostgreSQL

Haunted Escape (GameJam) **Nov 2024**

- Programmed core mechanics, physics systems, and UI/UX in Unity (C#) to deliver a 15-level 2D escape game during a GameJam, winning the Best Visual Award (£100).

Skills

- **AI & Research:** Python, PyTorch, Deep Learning, Computer Vision
- **Software Development:** C, C++, C#, Java, TypeScript, JavaScript
- **Databases & Infrastructure:** SQL, PostgreSQL, Supabase, Firebase
- **Tools & Frameworks:** Unity, Blender, React, Flutter, WinForms (.NET), Figma, Git